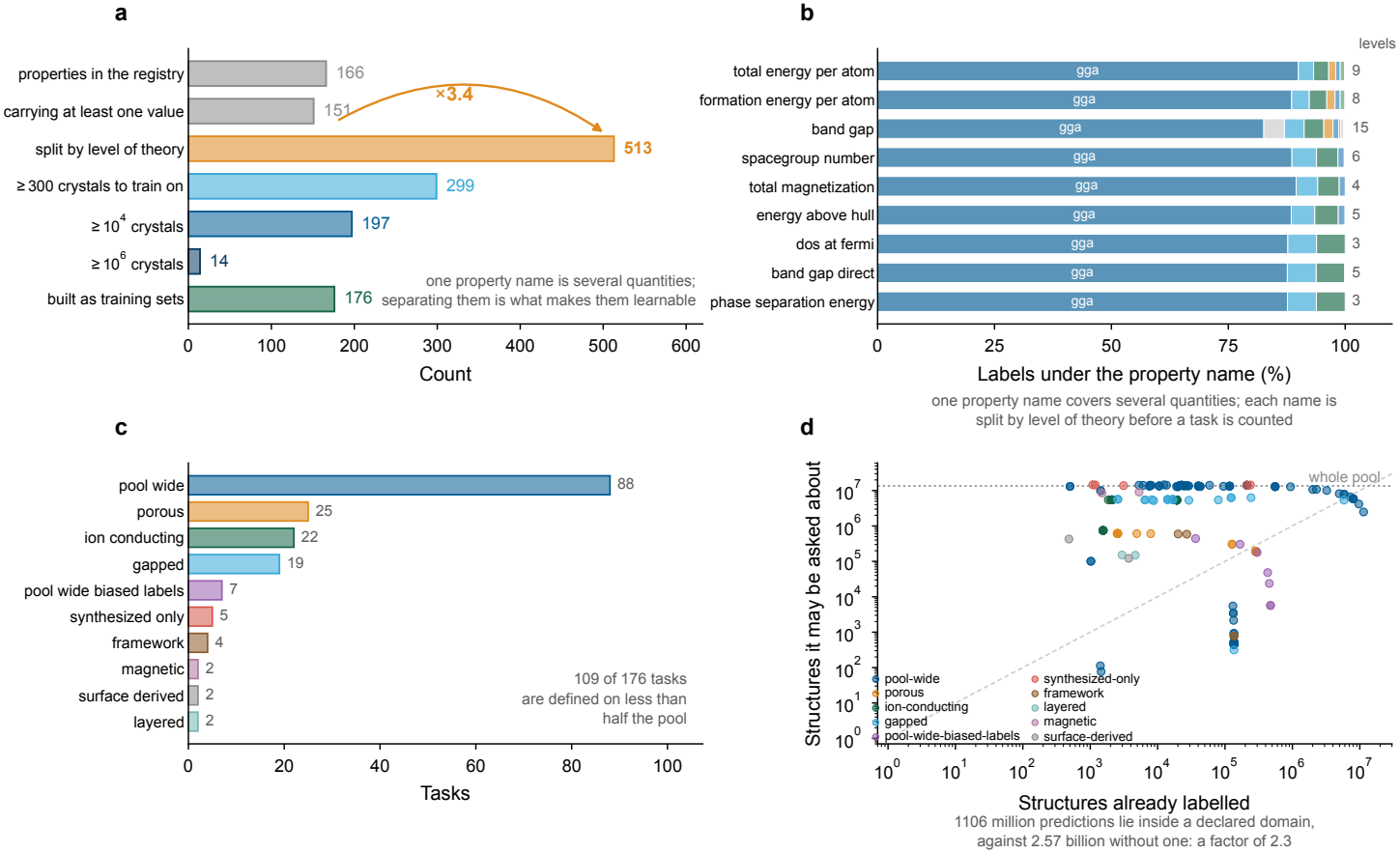
a
properties in the registry
166
carrying at least one value
151
×3.4
split by level of theory
513
≥ 300 crystals to train on
299
≥ 10^4 crystals
197
≥ 10^6 crystals
14
built as training sets
176
one property name is several quantities;
separating them is what makes them learnable
0
100
200
300
400
500
600
Count
b
levels
total energy per atom
gga
9
formation energy per atom
gga
8
band gap
gga
15
spacegroup number
gga
6
total magnetization
gga
4
energy above hull
gga
5
dos at fermi
gga
3
band gap direct
gga
5
phase separation energy
gga
3
0
25
50
75
100
Labels under the property name (%)
one property name covers several quantities; each name is
split by level of theory before a task is counted
c
pool wide
88
porous
25
ion conducting
22
gapped
19
pool wide biased labels
7
synthesized only
5
framework
4
magnetic
2
surface derived
2
layered
2
109 of 176 tasks
are defined on less than
half the pool
0
20
40
60
80
100
Tasks
d
Structures it may be asked about
whole pool
10^0
10^1
10^2
10^3
10^4
10^5
10^6
10^7
pool-wide
porous
ion-conducting
gapped
pool-wide-biased-labels
synthesized-only
framework
layered
magnetic
surface-derived
Structures already labelled
1106 million predictions lie inside a declared domain,
against 2.57 billion without one: a factor of 2.3